Blood group O people have no antigens, but both anti-A and anti-B antibodies in the plasma. Blood group AB people have both A and B antigens, but no antibodies

Rh System

Another group of antigens found on the red blood cells of people is the Rh factor (Rhesus monkey, in which these antigens were first discovered). The Rh blood group system was discovered in 1940 by Karl Landsteiner and A.S. Weiner. It is the second most important blood group system, after the ABO blood group system. There are five different antigens (C, D, E, c and e) in this group, but D antigen predominantly presents and is medically important. Thus Rh antigen is always referred to as the Antigen-D. If this Rh antigen is present on a person's red blood cells, the person is Rh positive; if it is absent, the person is Rh negative.

Thus the common major blood group systems are of eight blood groups:

A RhD positive (A+)

A RhD negative (A-)

B RhD positive (B+)

B RhD negative (B-)

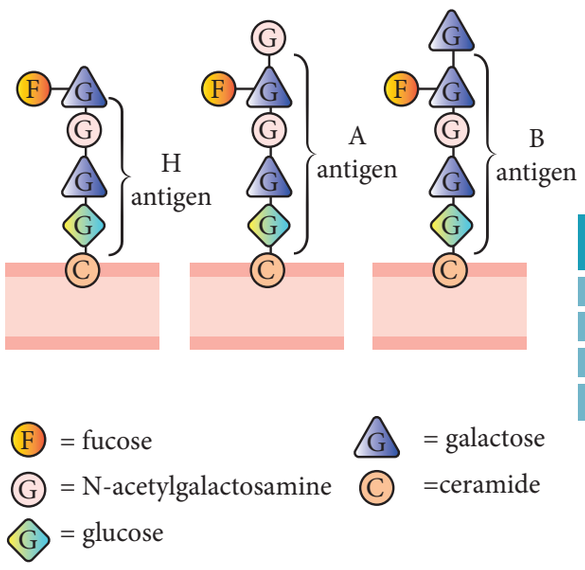
O RhD positive (O+)

O RhD negative (O-)

AB RhD positive (AB+)

AB RhD negative (AB-)

10.7.2. Antigens and Natural antibodies of ABO blood groups



Genotype	Antigen on RBCs	Antibody in Plasma
I^AI^A, I^Ai	A	Anti-B
I^BI^B, I^Bi	B	Anti-A
ii	O	Anti-A and anti-B
I^AI^B	AB	Neither anti-A and anti-B

XII U10 Immunology.indd 254 4/8/2020 3:28:03 PM